SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing techniques. (BL4,BL5)
Assessment Tool Weightage: 40%

QUESTIONS: 05 **Total Marks:40**

Annexure A

MOCK INTERVIEW

Code of Conduct:

I _E. Sakthi priya (192520041)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Sakthi priya

Annexure A

Mock Interview – Sample Questions (5 Questions)

Mock Interview Test Format for Object Oriented Analysis and Design (OOAD)

Section A	**Self Introduction & Fundamentals**	**3 Minutes**	**5**
Section B	**Technical Questions (OOAD Concepts)**	**7 Minutes**	**10**
Section C	**UML Modeling & Design**	**7 Minutes**	**10**
Section D	**Case Study / Problem Solving**	**8 Minutes**	**10**
Section E	**HR & Career-Oriented Questions**	**5 Minutes**	**5**

SECTION A – SELF INTRODUCTION (5 Marks)

Evaluation Criteria

- Communication Skills
- Confidence
- Technical Awareness
- Academic Background

Sample Questions

1. Tell me about yourself.
2. Why did you choose Computer Science?
3. Explain your favorite subject in OOAD.
4. What software development tools have you used?
5. Describe a project where you applied object-oriented concepts.

SECTION B – TECHNICAL QUESTIONS (15 Marks)

Basic Level

Q1. What is Object-Oriented Analysis and Design?

Q2. Differentiate between Analysis and Design.

Q3. Explain Encapsulation with an example.

Q4. What is Inheritance? Mention its types.

Q5. Define Polymorphism and its benefits.

Q6. What is Abstraction?

Q7. Difference between Aggregation and Composition.

Q8. What is Coupling and Cohesion?

Q9. What is a Class and an Object?

Q10. Explain Object Responsibility.

SECTION C – UML MODELING (15 Marks)

Q11. What is UML?

Q12. When would you use a Use Case Diagram?

Q13. Draw and explain a Class Diagram for a Library Management System.

Q14. Differentiate Sequence Diagram and Activity Diagram.

Q15. Explain State Diagram with an example.

Q16. What is a Deployment Diagram?

Q17. How do Component Diagrams help software architects?

Q18. What is Multiplicity in UML?

SECTION D – CASE STUDY / PROBLEM SOLVING (10 Marks)

Case Study 1

A hospital wants a system to manage:

- Patients
- Doctors
- Appointments
- Billing

Questions

1. Identify the classes.
2. Draw a simple class diagram.
3. Suggest relationships among classes.
4. Which UML diagrams would you use for analysis?
5. Explain how inheritance can be applied.

Case Study 2

An Online Shopping System supports:

- Customers
- Products
- Cart
- Orders
- Payments

Questions

1. Identify actors and use cases.
2. Design a use case diagram.
3. Suggest suitable design patterns.
4. Explain how polymorphism can be used in payment processing.

SECTION E – HR & PROFESSIONAL QUESTIONS (5 Marks)

Q19. Why should we hire you?

Q20. What are your strengths and weaknesses?

Q21. How do you handle project deadlines?

Q22. Have you worked in a team project? Explain your role.

Q23. Where do you see yourself in five years?

RUBRICS FOR CALCULATION:

MOCK INTERVIEW

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

SECTION A – SELF INTRODUCTION

1. Tell me about yourself.

I am a second-year Bioinformatics student interested in programming, software design, and computational biology. I am currently learning OOAD, programming, and bioinformatics tools.

2. Why did you choose Computer Science?

Computer science helps me develop programming and problem-solving skills, which are useful in bioinformatics and healthcare applications.

3. Favorite subject in OOAD?

My favorite topic is UML, because diagrams make software structure and interactions easy to understand.

4. What software tools have you used?

I have used PlantUML, Visual Studio Code, Google Colab, and various bioinformatics tools.

5. Project using OOP concepts?

A Library Management System can use classes such as Book, Member, and Librarian, with concepts like encapsulation and inheritance.

SECTION B – TECHNICAL QUESTIONS

The assessment includes 10 basic OOAD questions.

Q1. What is OOAD?

OOAD means Object-Oriented Analysis and Design. It analyzes and designs software using objects, classes, and their relationships.

Q2. Analysis vs Design?

Analysis: What the system should do.

Design: How the system will do it.

Q3. Encapsulation?

Wrapping data and methods inside a class and restricting direct access to data.

Example: Private balance in a BankAccount class.

Q4. Inheritance?

A child class gets properties and methods from a parent class.

Types: Single, Multilevel, Hierarchical, Multiple, Hybrid.

Q5. Polymorphism?

One interface or method can have different implementations.

Example: pay() can work differently for UPI and Credit Card.

Q6. Abstraction?

Hiding unnecessary implementation details and showing only essential features.

Q7. Aggregation vs Composition?

Aggregation: Weak relationship; objects can exist independently.

Composition: Strong relationship; child depends on parent.

Q8. Coupling and Cohesion?

Coupling: Dependency between classes.

Cohesion: How closely related the responsibilities of a class are.

Good design = Low Coupling + High Cohesion.

Q9. Class vs Object?

Class: Blueprint.

Object: Instance of a class.

Q10. Object Responsibility?

The specific task or information an object is responsible for handling.

SECTION C – UML

The assessment asks about UML, use cases, class diagrams, sequence/activity diagrams, state, deployment, components, and multiplicity.

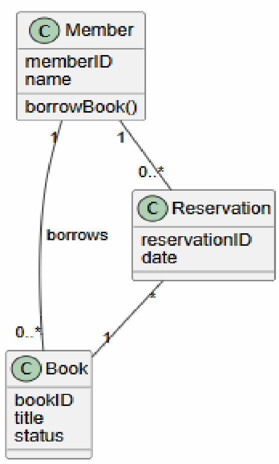
Q11. What is UML?

Unified Modeling Language is used to visualize and document software systems.

Q12. Use Case Diagram?

Shows actors, system functions, and their interactions.

Q13. Library Class Diagram



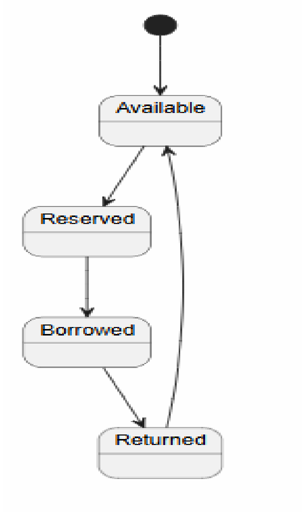
Q14. Sequence vs Activity Diagram?

Sequence: Shows object interactions in time order.

Activity: Shows workflow/process.

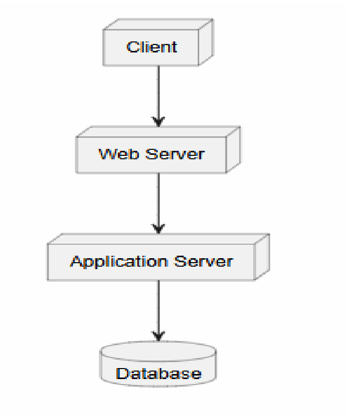
Q15. State Diagram?

Shows different states of an object.



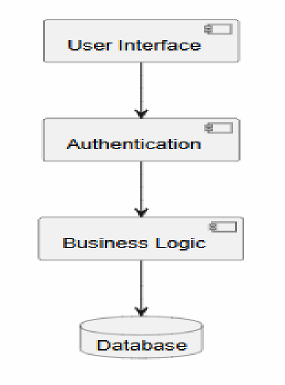
Q16. Deployment Diagram?

Shows how software is deployed on physical devices/servers.



Q17. Component Diagram?

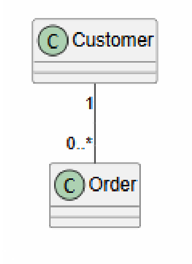
Shows major software components and their dependencies.



Q18. Multiplicity?

Shows the number of objects participating in a relationship.

Example: One customer can have many orders.



SECTION D

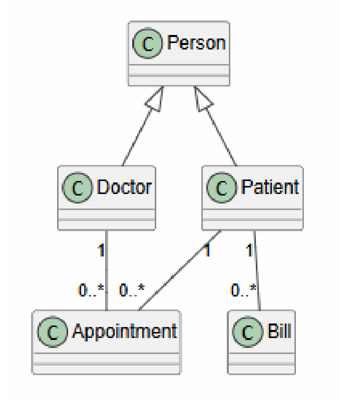
CASE STUDY 1: HOSPITAL

The hospital system contains Patients, Doctors, Appointments, and Billing.

1. Identify classes

Patient, Doctor, Appointment, Bill

2. Class Diagram



3. Relationships

- Patient books Appointment.
- Doctor attends Appointment.
- Patient has Bill.

4. UML diagrams for analysis

Use Case, Class, Sequence, Activity, and State diagrams.

5. Inheritance

Person can be the parent class, while Patient and Doctor inherit common properties such as name and age.

CASE STUDY 2 – ONLINE SHOPPING

The system includes Customers, Products, Cart, Orders, and Payments.

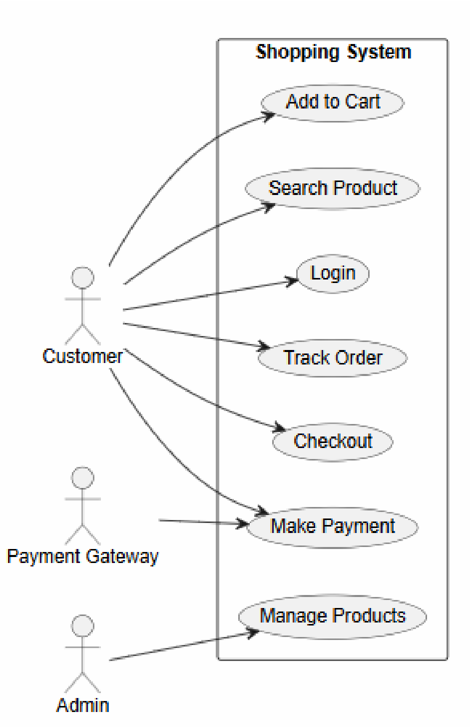
1. Actors

- Customer
- Admin
- Payment Gateway

Use Cases

- Login
- Search Product
- Add to Cart
- Checkout
- Make Payment
- Track Order
- Manage Products

2. Use Case Diagram



3. Suitable Design Patterns

- Strategy: Different payment methods.
- Factory: Create payment objects.
- Observer: Order/delivery notifications.

4. Polymorphism in Payment

A common pay() method can be implemented differently by UPI, Credit Card, and Net Banking classes.

SECTION E – HR QUESTIONS

The assessment includes five HR questions.

Q19. Why should we hire you?

I am a quick learner, responsible, and willing to improve my technical skills. I can adapt to new technologies and work sincerely on assigned tasks.

Q20. Strengths and weaknesses?

Strengths: Quick learner, hardworking, adaptable.

Weakness: I sometimes take extra time to understand new concepts, but I overcome this through practice.

Q21. How do you handle deadlines?

I divide the work into smaller tasks, prioritize important work, and track my progress to complete everything on time.

Q22. Team project and your role?

Yes. I participate in discussions, complete my assigned tasks, communicate with teammates, and help combine the final work.

Q23. Where do you see yourself in five years?

I want to have strong skills in bioinformatics, programming, and computational analysis and work on real-world biological or healthcare problems.